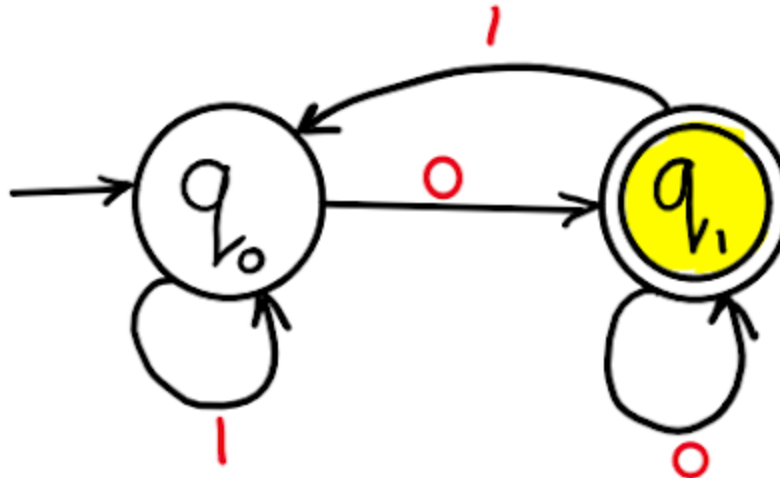


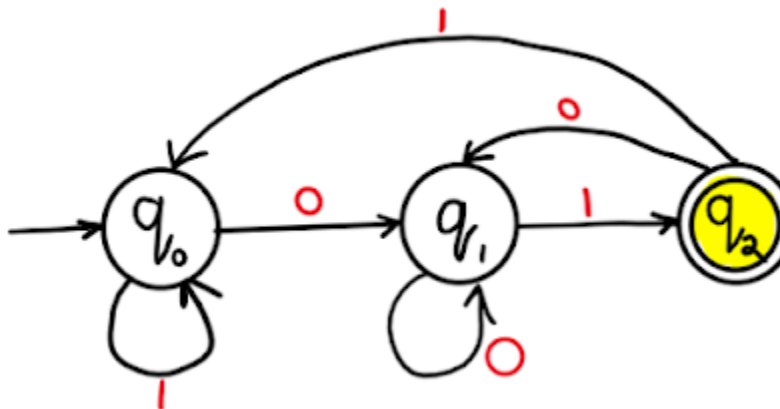
Example 1: Draw a DFA for the language accepting strings ending with '0' over input alphabets $\Sigma=\{0, 1\}$?

Solution:



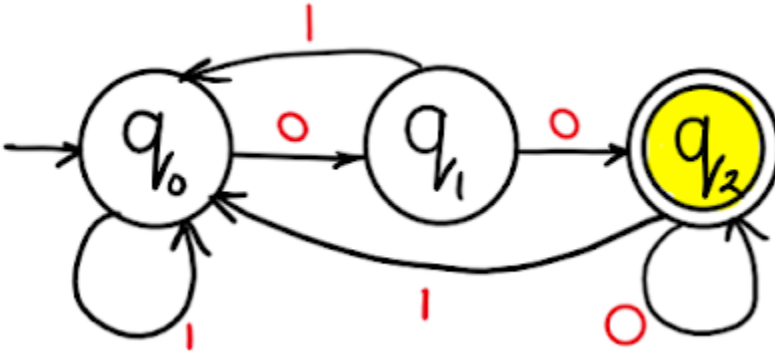
Example 2: Draw a DFA for the language accepting strings ending with '01' over input alphabets $\Sigma=\{0, 1\}$?

Solution:



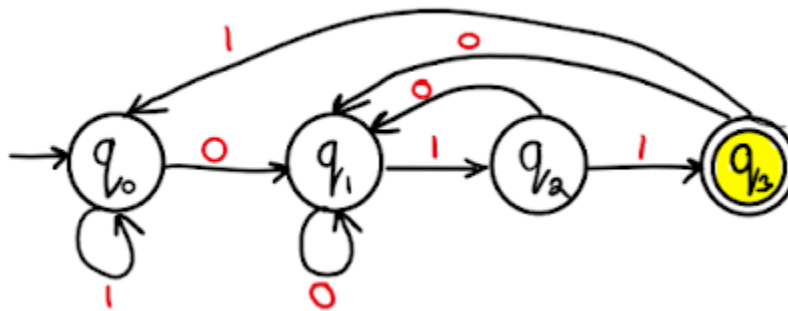
Example 3: Draw a DFA for the language accepting strings ending with '00' over input alphabets $\Sigma=\{0, 1\}$?

Solution:



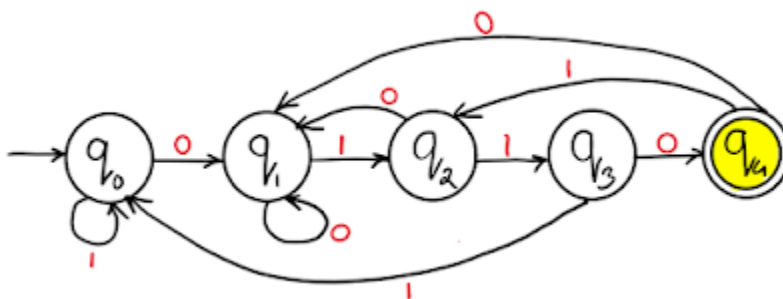
Example 4: Draw a DFA for the language accepting strings ending with '011' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



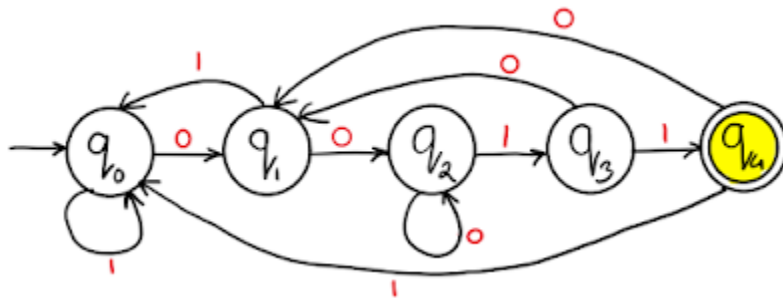
Example 5: Draw a DFA for the language accepting strings ending with '0110' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



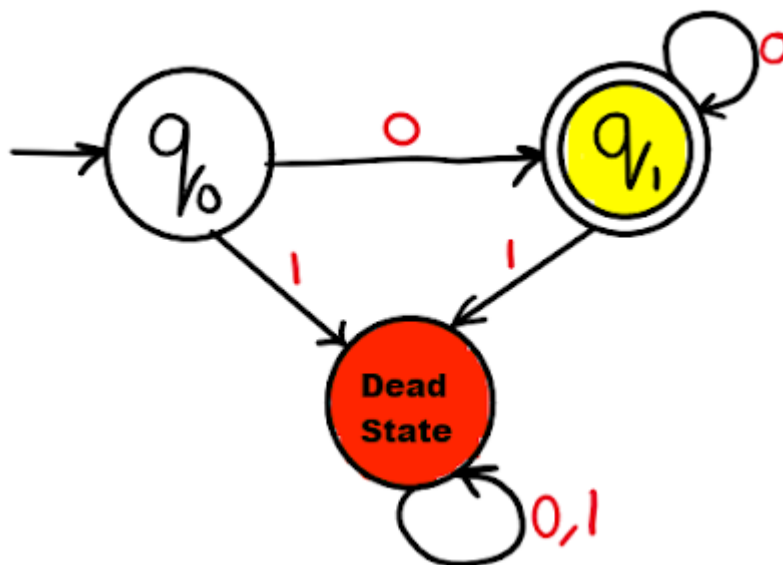
Example 6: Draw a DFA for the language accepting strings ending with '0011' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



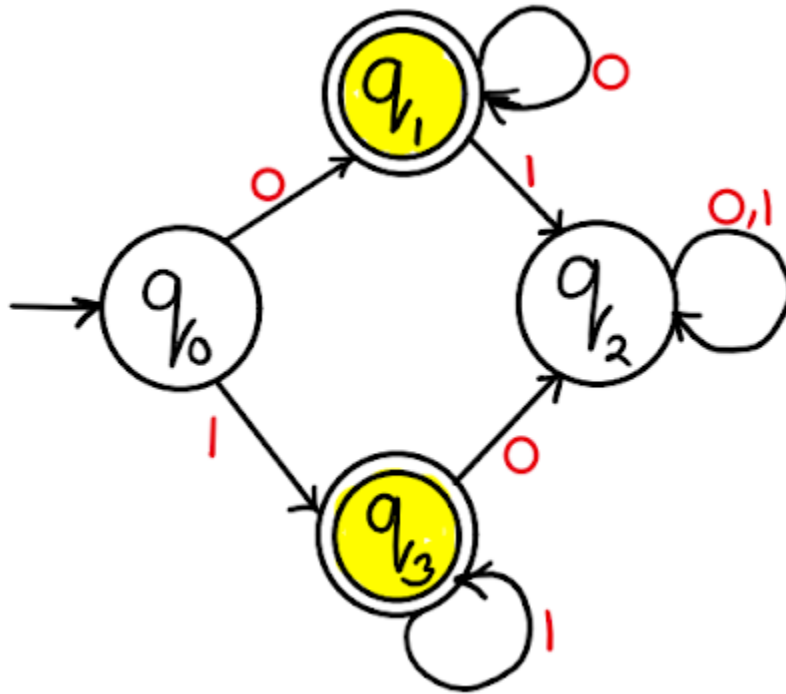
Example 7: Draw a DFA for the language accepting strings with '0' only over input alphabets $\Sigma=\{0, 1\}$?

Solution:



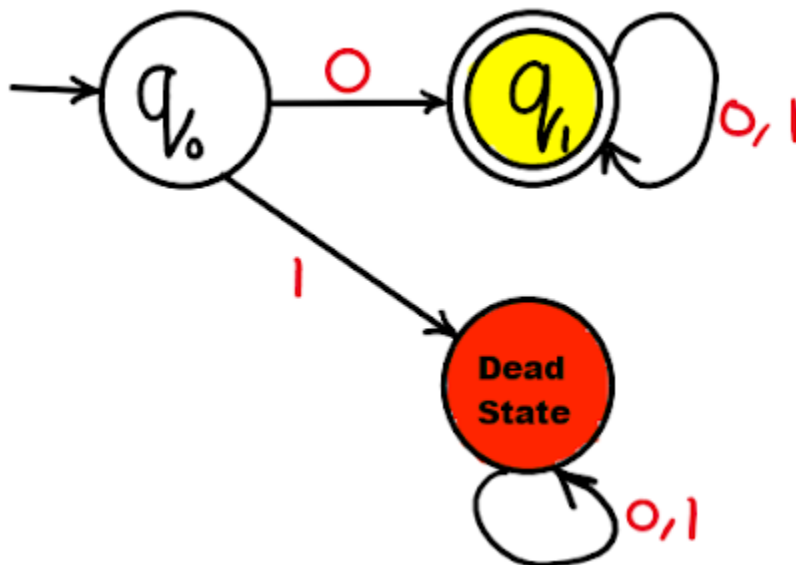
Example 8: Draw a DFA for the language accepting strings with '0' and '1' only over input alphabets $\Sigma=\{0, 1\}$?

Solution:



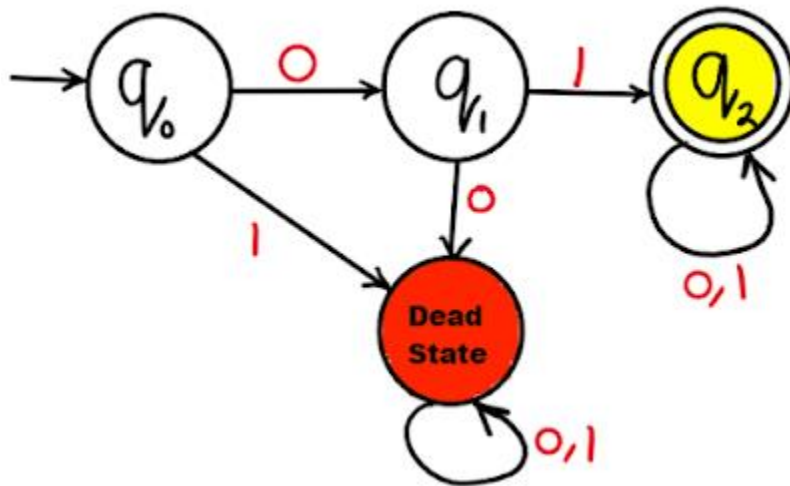
Example 9: Draw a DFA for the language accepting strings starting with '0' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



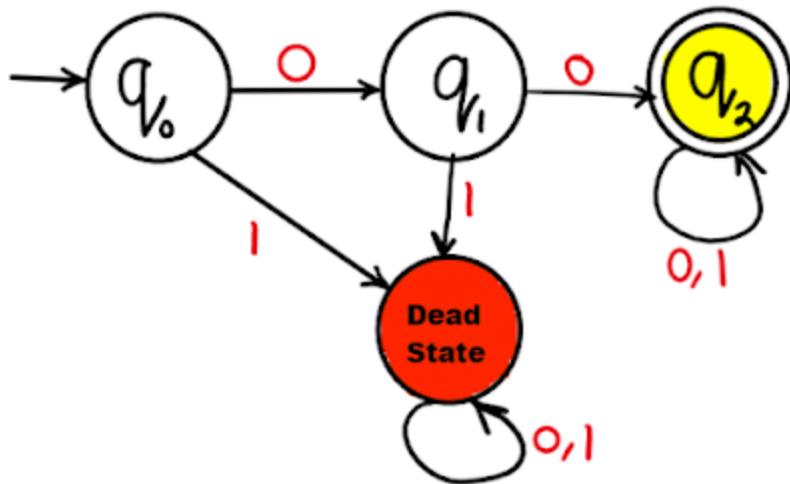
Example 10: Draw a DFA for the language accepting strings starting with '01' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



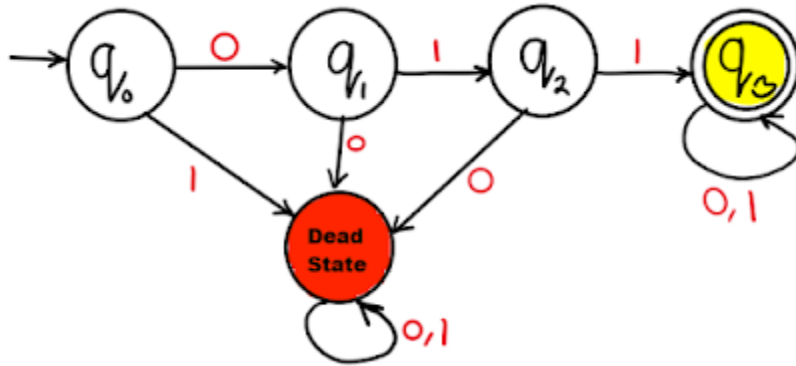
Example 11: Draw a DFA for the language accepting strings starting with '00' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



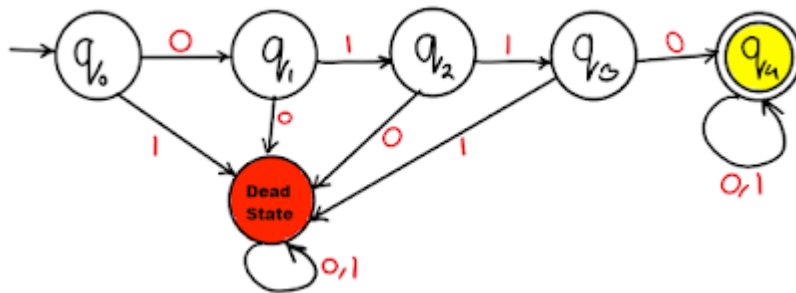
Example 12: Draw a DFA for the language accepting strings starting with '011' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



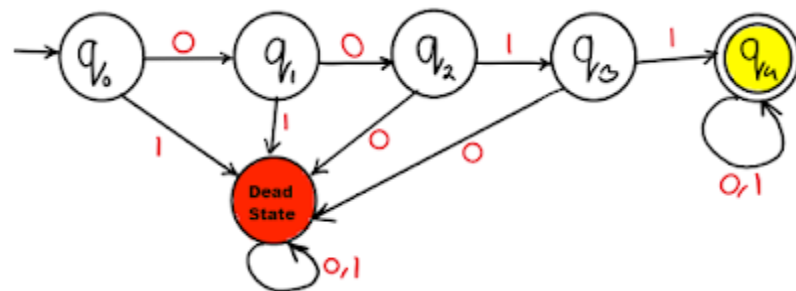
Example 13: Draw a DFA for the language accepting strings starting with '0110' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



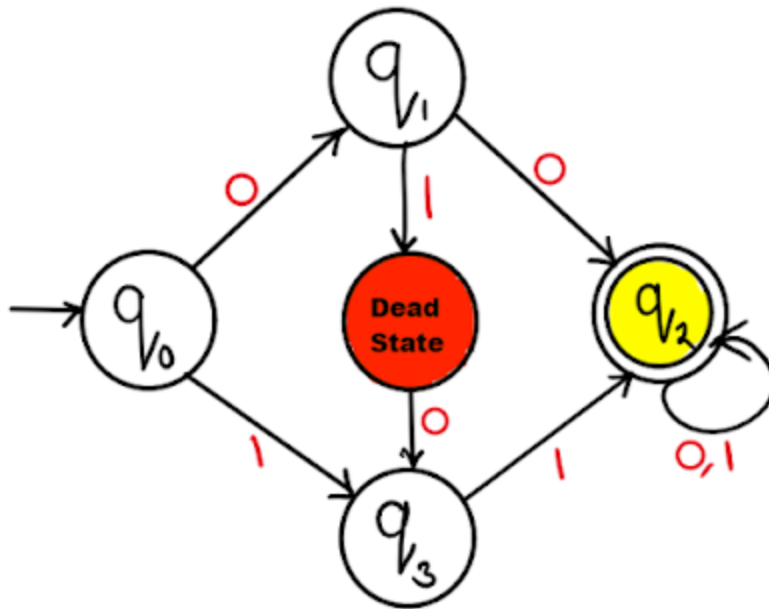
Example 14: Draw a DFA for the language accepting strings starting with '0011' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



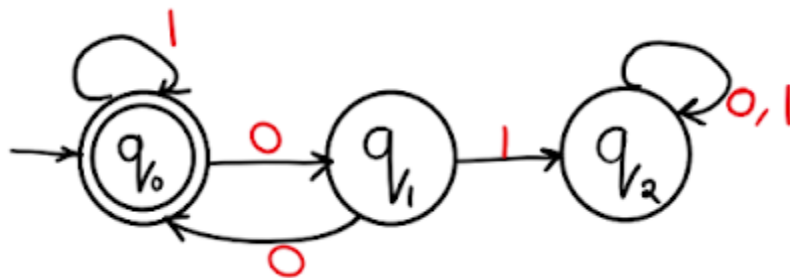
Example 15: Draw a DFA for the language accepting strings starting with '00' or '11' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



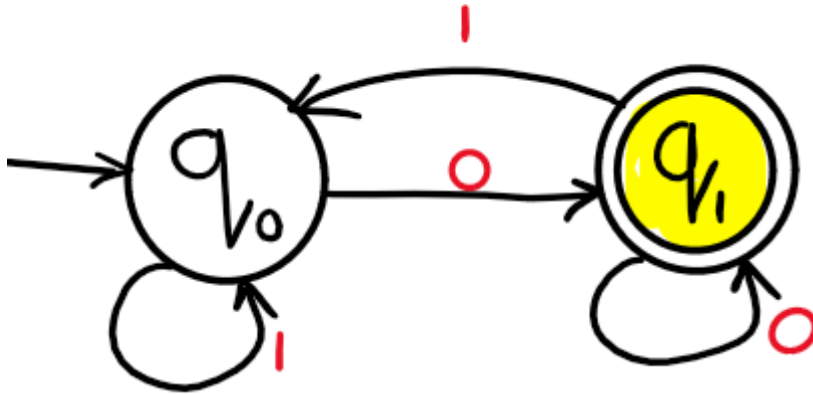
Example 16: Draw a DFA for the language accepting strings without substring '00' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



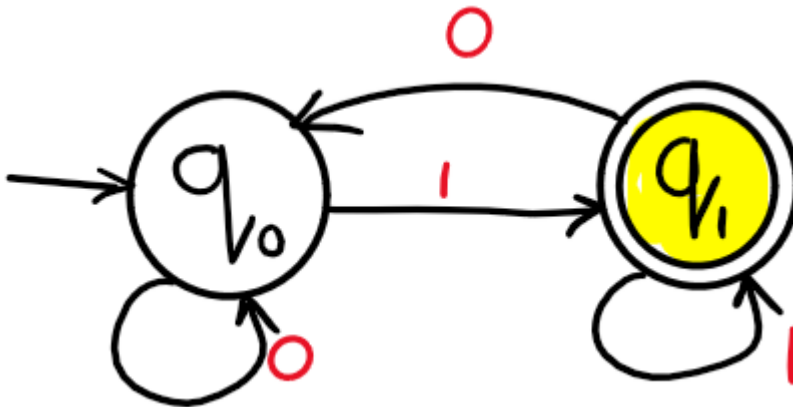
Example 17: Draw a DFA for the language accepting even binary numbers strings over input alphabets $\Sigma = \{0, 1\}$?

Soluntion:



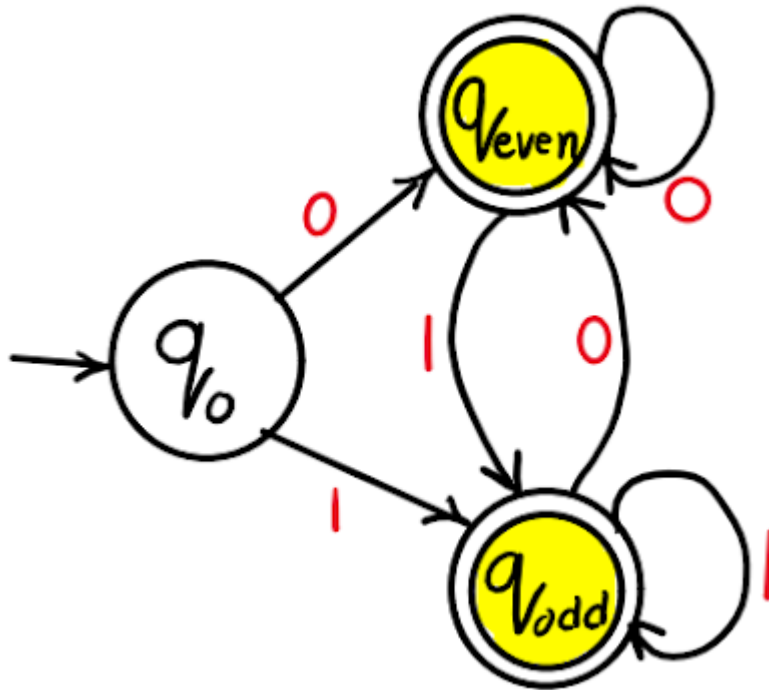
Example 18: Draw a DFA for the language accepting odd binary numbers strings over input alphabets $\Sigma = \{0, 1\}$?

Solution:



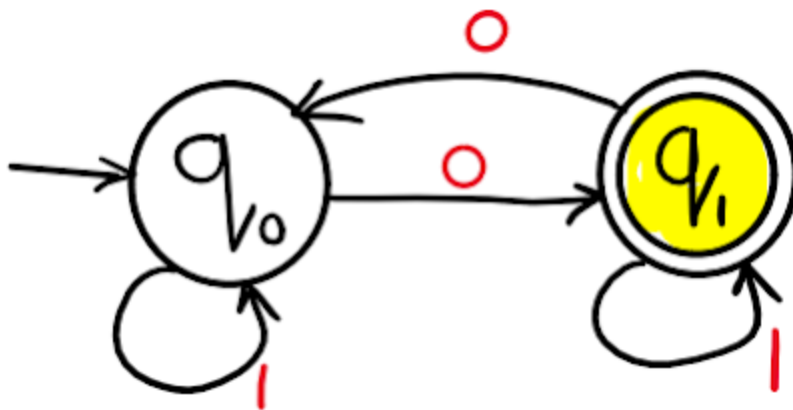
Example 19: Draw a DFA for the language accepting odd or even binary numbers strings over input alphabets $\Sigma = \{0, 1\}$?

Solution:



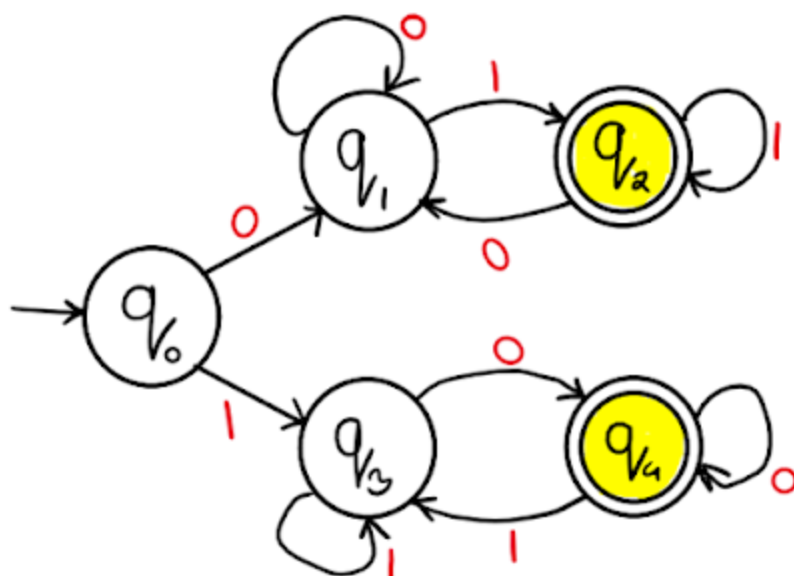
Example 20: Draw a DFA for the language accepting strings containing even number of total zeros over input alphabets $\Sigma = \{0, 1\}$?

Solution:



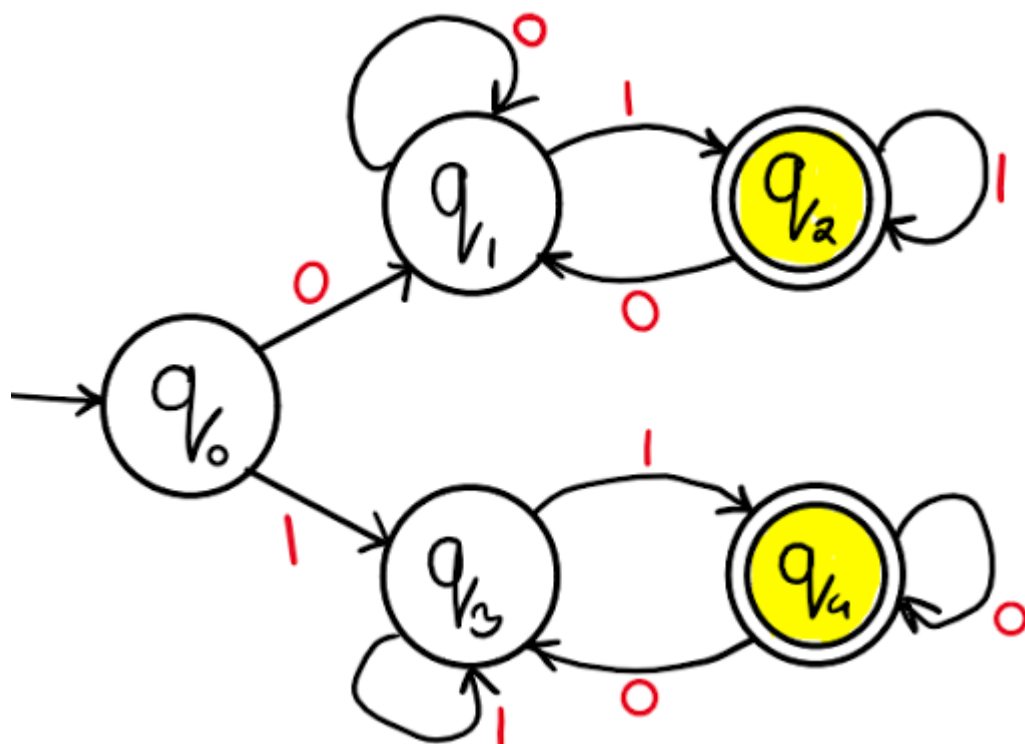
Example 21: Draw a DFA for the language accepting strings starting and ending with different characters over input alphabets $\Sigma = \{0, 1\}$?

Solution:



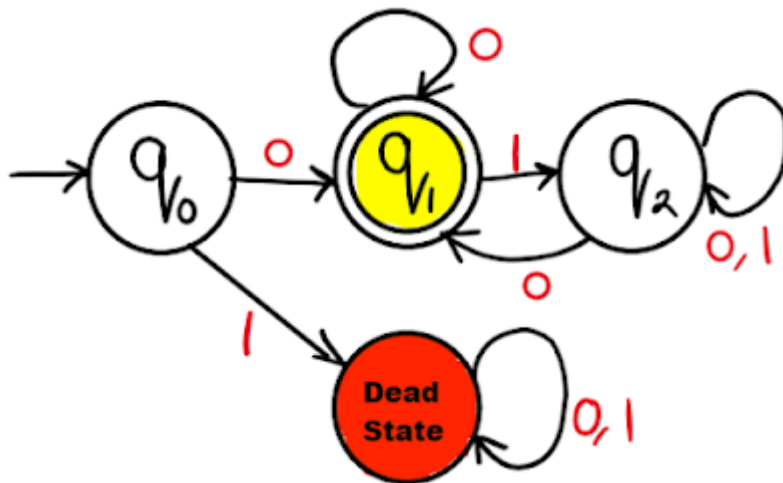
Example 22: Draw a DFA for the language accepting strings starting and ending with same character over input alphabets $\Sigma = \{0, 1\}$?

Solution:



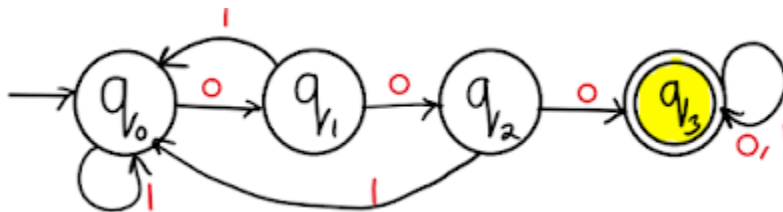
Example 23: Draw a DFA for the language accepting strings starting and ending with '0' always over input alphabets $\Sigma = \{0, 1\}$?

Solution:



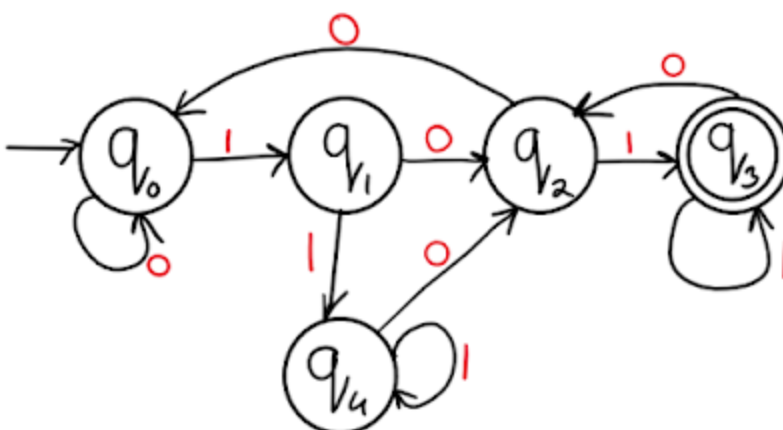
Example 24: Draw a DFA for the language accepting strings containing three consecutive '0' always over input alphabets $\Sigma = \{0, 1\}$?

Solution:



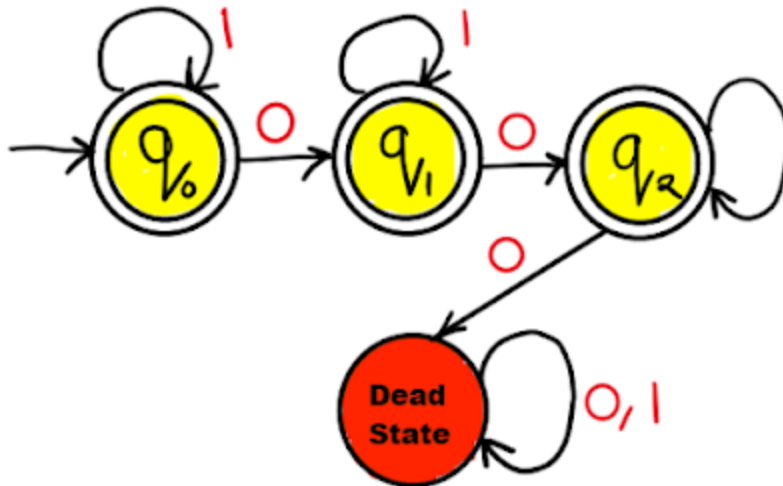
Example 25: Draw a DFA for the language accepting strings such that each '0' is immediately preceded and followed by '1' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



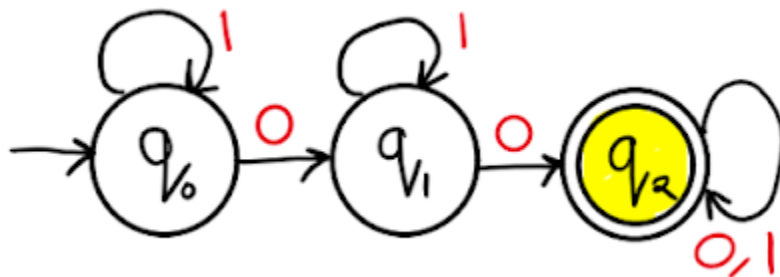
Example 26: Draw a DFA for the language accepting strings containing at most two '0' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



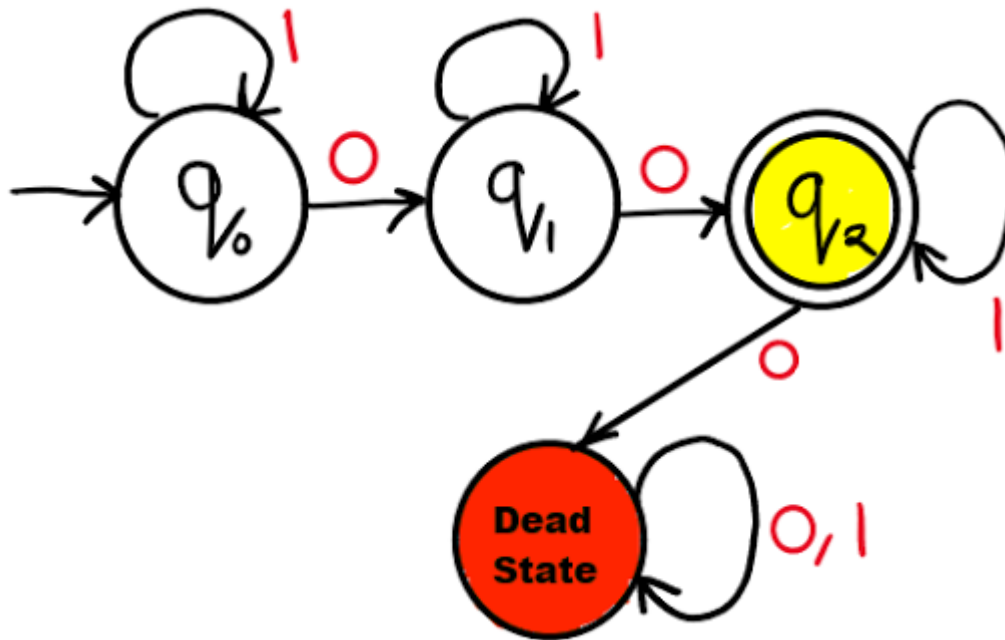
Example 27: Draw a DFA for the language accepting strings containing at least two '0' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



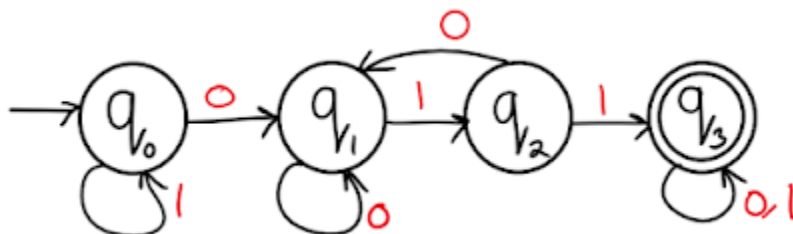
Example 28: Draw a DFA for the language accepting strings containing exactly two '0' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



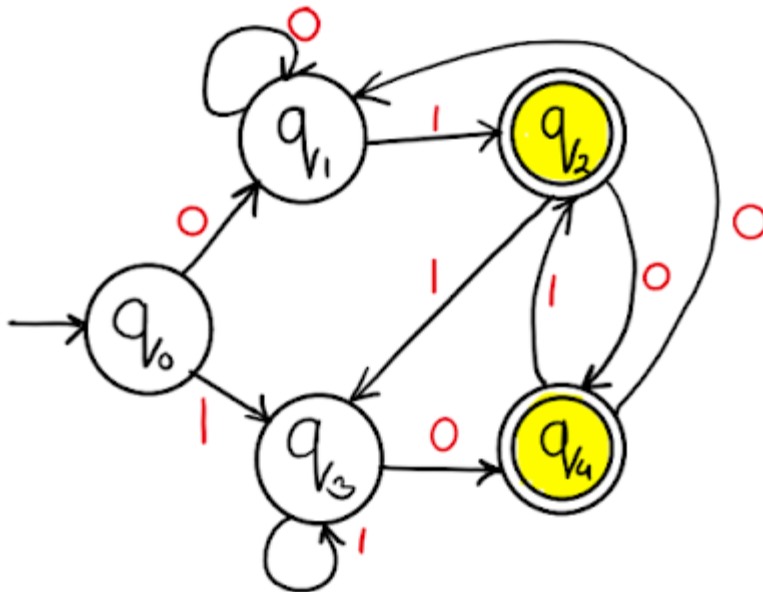
Example 29: Draw a DFA for the language accepting strings with '011' as substring over input alphabets $\Sigma = \{0, 1\}$?

Solution:



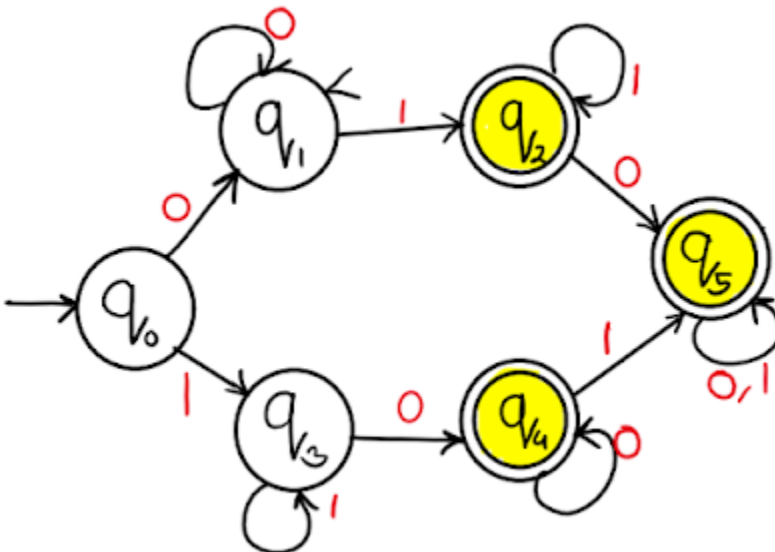
Example 30: Draw a DFA for the language accepting strings ending in either '01', or '10' over input alphabets $\Sigma = \{0, 1\}$?

Solution:



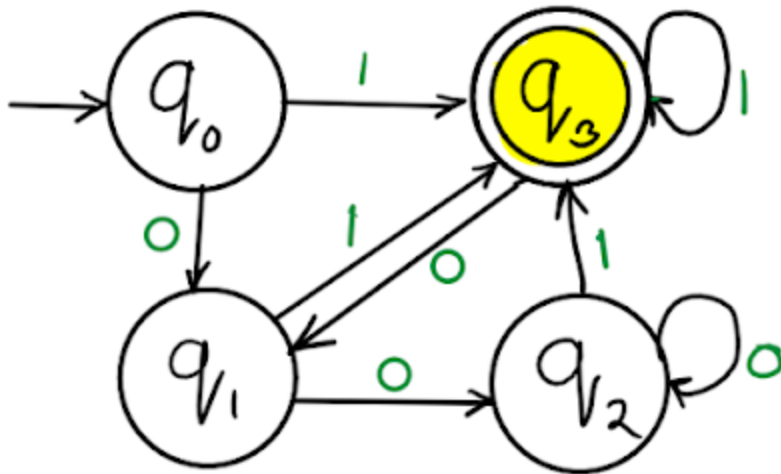
Example 31: Draw a DFA for the language accepting strings containing '01', or '10' as substring over input alphabets $\Sigma = \{0, 1\}$?

Solution:



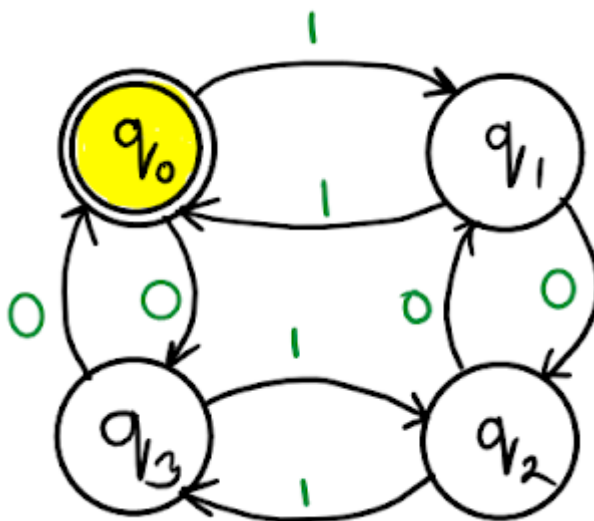
Example 32: Draw DFA that accepts any string which ends with 1 or it ends with an even number of 0's following the last 1. Alphabets are $\{0,1\}$.

Solution:



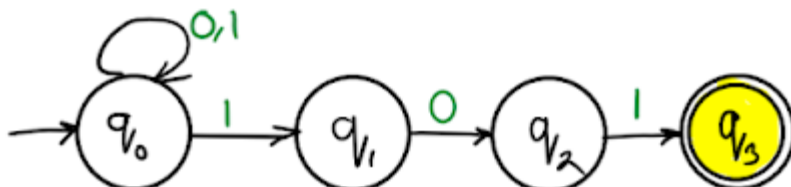
Example 33: Construct DFA accepting set of all strings containing even no. of a's and even no. of b's over input alphabet $\{a,b\}$.

Solution:



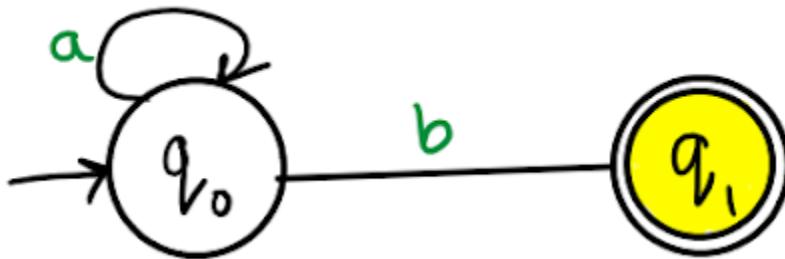
Example 34: Give DFA accepting the language over alphabet $\{0,1\}$ such that all strings of 0 and 1 ending in 101.

Solution:



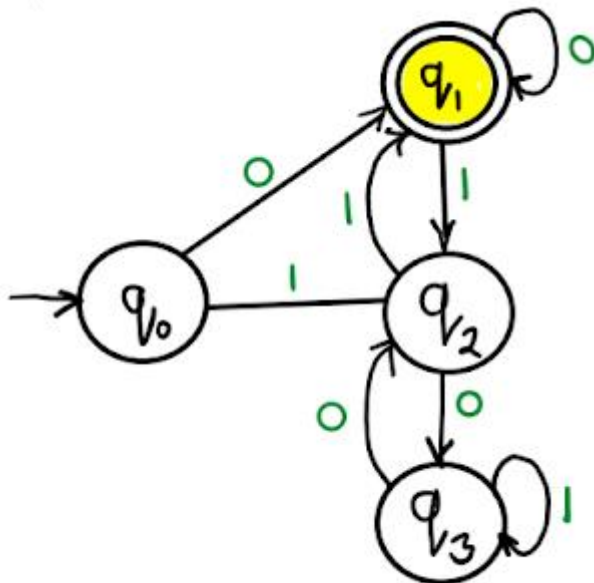
Example 35: Construct DFA for $anb \mid n \geq 0$.

Solution:



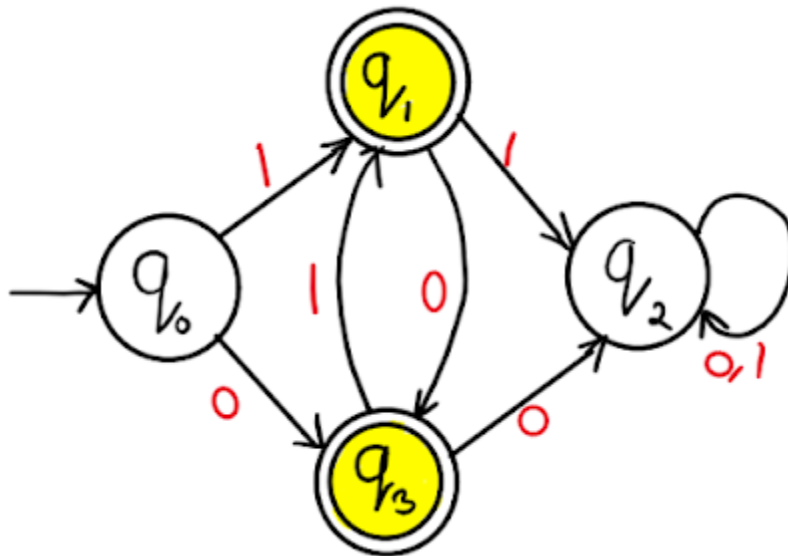
Example 36: construct DFA for binary integer divisible by 3 ?

Solution:



Example 37: Draw a DFA for the language accepting strings containing neither '00', nor '11' as substring over input alphabets $\Sigma = \{0, 1\}$?

Solution:



Problem-01:

Draw a DFA for the language accepting strings starting with 'ab' over input alphabets $\Sigma = \{a, b\}$

Solution-

Regular expression for the given language = $ab(a + b)^*$

Step-01:

- All strings of the language starts with substring "ab".
- So, length of substring = 2.

Thus, Minimum number of states required in the DFA = $2 + 2 = 4$.

It suggests that minimized DFA will have 4 states.

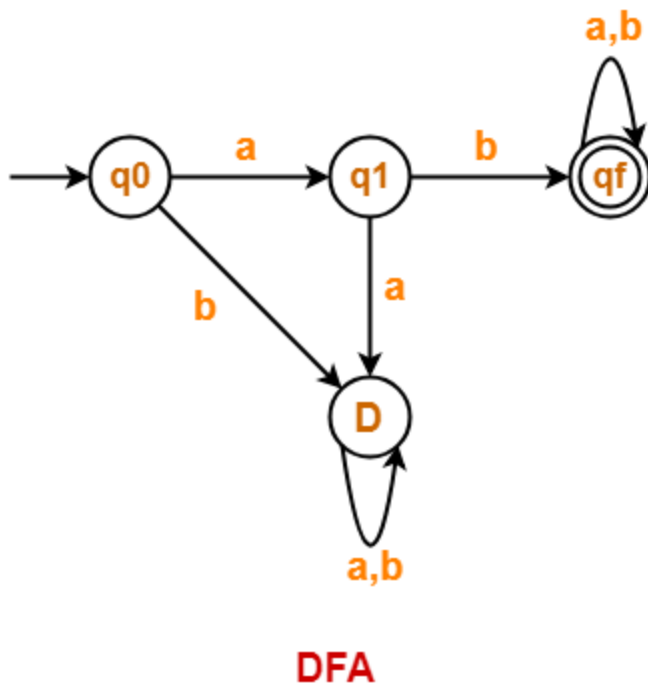
Step-02:

We will construct DFA for the following strings-

- ab
- aba
- abab

Step-03:

The required DFA is-



Problem-02:

Draw a DFA for the language accepting strings starting with 'a' over input alphabets $\Sigma = \{a, b\}$

Solution-

Regular expression for the given language = $a(a + b)^*$

Step-01:

- All strings of the language starts with substring “a”.
- So, length of substring = 1.

Thus, Minimum number of states required in the DFA = $1 + 2 = 3$.

It suggests that minimized DFA will have 3 states.

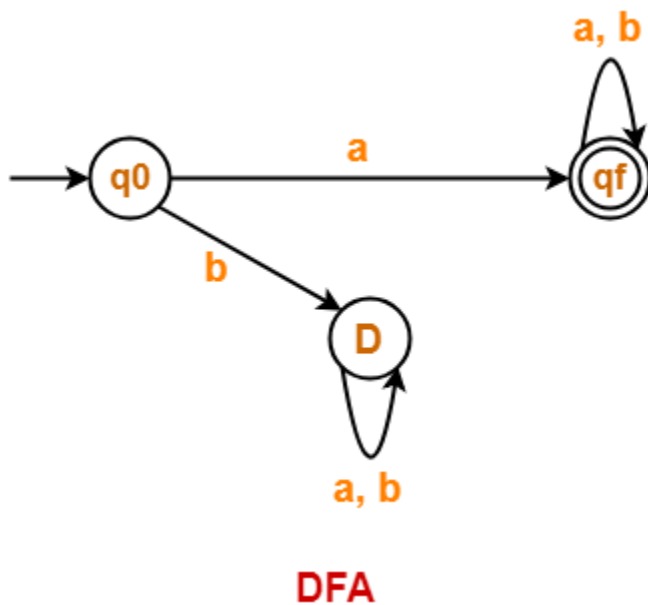
Step-02:

We will construct DFA for the following strings-

- a
- aa

Step-03:

The required DFA is-



Problem-03:

Draw a DFA for the language accepting strings starting with '101' over input alphabets $\Sigma = \{0, 1\}$

Solution-

Regular expression for the given language = $101(0 + 1)^*$

Step-01:

- All strings of the language starts with substring "101".
- So, length of substring = 3.

Thus, Minimum number of states required in the DFA = $3 + 2 = 5$.

It suggests that minimized DFA will have 5 states.

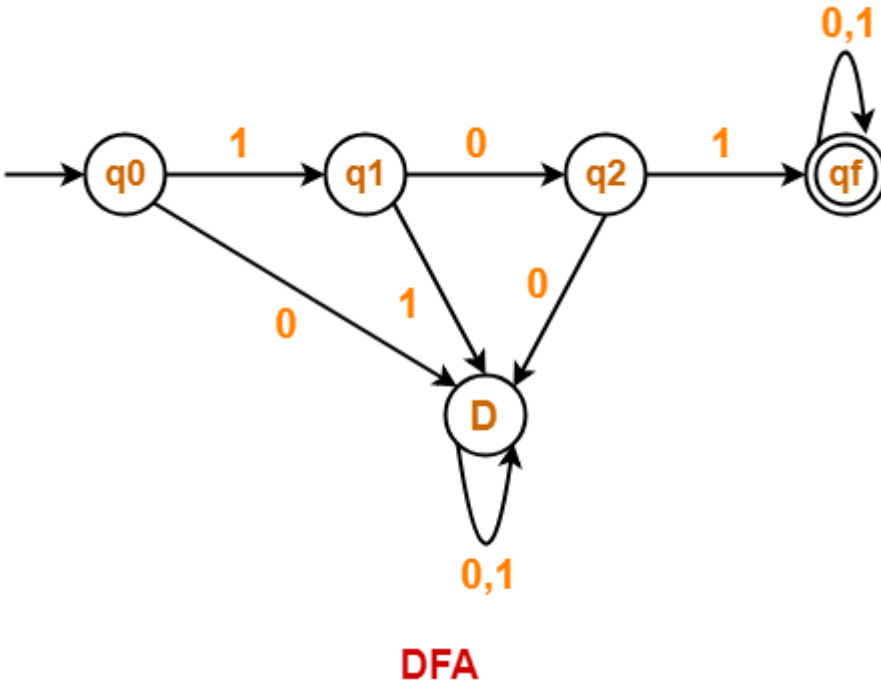
Step-02:

We will construct DFA for the following strings-

- 101
- 1011
- 10110
- 101101

Step-03:

The required DFA is-



Problem-04:

Draw a DFA that accepts a language L over input alphabets $\Sigma = \{0, 1\}$ such that L is the set of all strings starting with '00'.

Solution-

Regular expression for the given language = $00(0 + 1)^*$

Step-01:

- All strings of the language starts with substring "00".
- So, length of substring = 2.

Thus, Minimum number of states required in the DFA = $2 + 2 = 4$.

It suggests that minimized DFA will have 4 states.

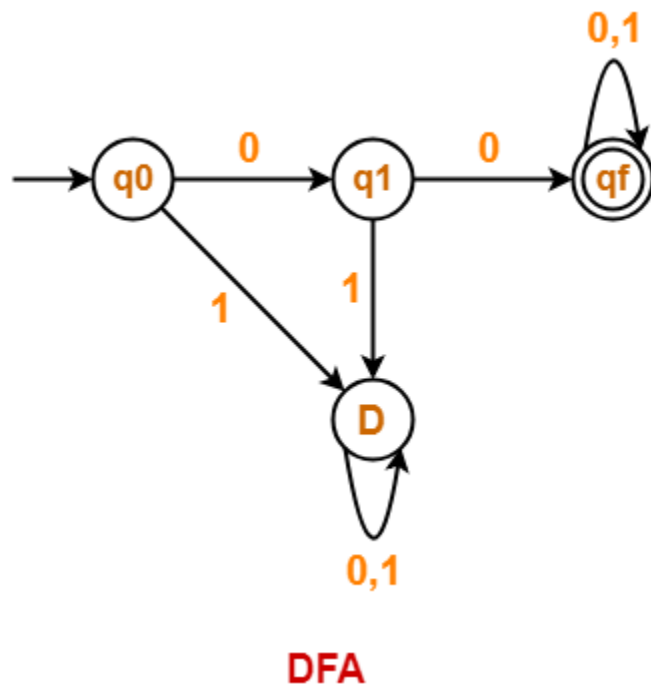
Step-02:

We will construct DFA for the following strings-

- 00
- 000
- 00000

Step-03:

The required DFA is-



Problem-05:

Construct a DFA that accepts a language L over input alphabets $\Sigma = \{a, b\}$ such that L is the set of all strings starting with 'aa' or 'bb'.

Solution-

Regular expression for the given language = $(aa + bb)(a + b)^*$

Step-01:

Minimum number of states required in the DFA = 5.

It suggests that minimized DFA will have 5 states.

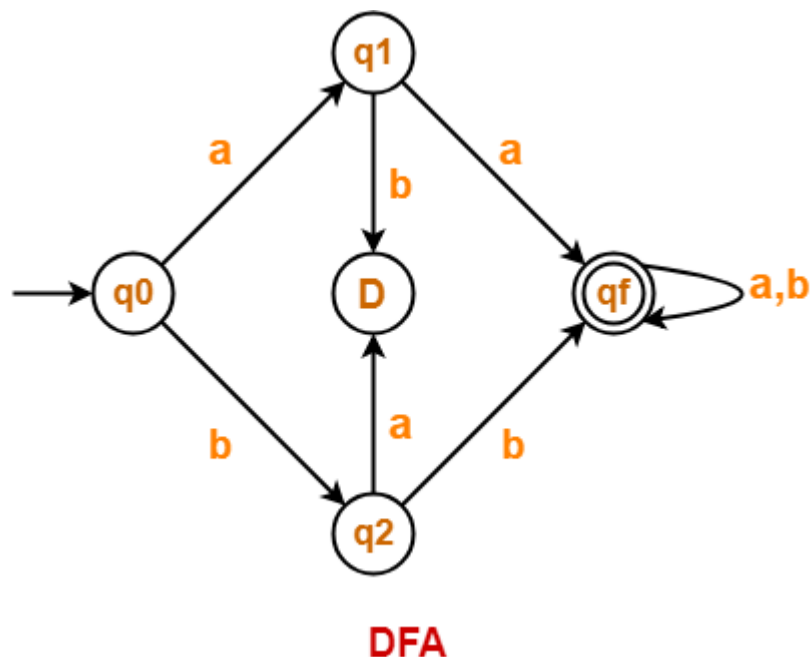
Step-02:

We will construct DFA for the following strings-

- aa
- aaa
- aaaa
- bb
- bbb
- bbbb

Step-03:

The required DFA is-



Problem-06:

Construct a DFA that accepts a language L over input alphabets $\Sigma = \{a, b\}$ such that L is the set of all strings starting with 'aba'.

Solution-

Regular expression for the given language = $aba(a + b)^*$

Step-01:

- All strings of the language starts with substring "aba".
- So, length of substring = 3.

Thus, Minimum number of states required in the DFA = $3 + 2 = 5$.

It suggests that minimized DFA will have 5 states.

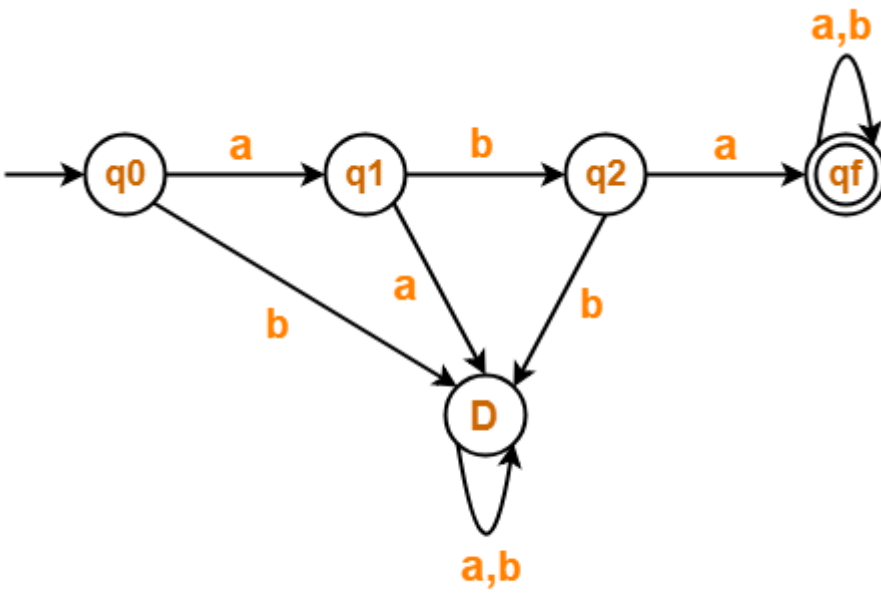
Step-02:

We will construct DFA for the following strings-

- aba
- abaa
- abaab
- abaaba

Step-03:

The required DFA is-



DFA